

AP 2015
Grade-10
General Science Paper-I
(English version)
Parts A and B

[Time: 2 hrs. 45 min]

[Maximum Marks: 50]

Instructions:

1. This paper contains Part-A
2. Answer the questions. under Part-A on the separate answer book. Answer all the questions: 'Internal choice is given to the questions under Section-III.
3. In the duration, of 2.45 hours, 15 minutes of time is allotted to read the question paper.

Part – A

Time: 2 Hours

Marks: 35

SECTION -I

5x2=10

Note:

1. Answer **ANY FIVE** questions, choosing at least **two** from each Group:
2. Each question carries **TWO** marks.

GROUP-A

1. State any two differences between Convex and Concave mirrors.

Ans:

Convex Mirror	Concave Mirror
Reflecting Surface is bulged outward	Reflecting Surface is bulged inward
Focal Point is behind the mirror	Focal Point is in front the mirror

2. Write the Lens maker's formula and explain the terms in it.

Ans: The lens maker formula gives the relationship between focal length, radius of curvature and the refractive index of the lens.

Mathematically it is given as:

$$\frac{1}{F} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

Where,

F is the focal length

n is the refractive index

R_1 , R_2 are the radius of curvature of the lens

3. What do you mean by electric shock? Explain how it takes place?

Ans: When there is a difference between electric current and resistance of human body, then we suffer electric shock.

So when the current flows via a human body then the path that it follows offers it low resistance so the body gets electric shock.

4. By observing steel vessels and different images in them, Ramu, a third-class student asked some questions to his elder sister Santha. What may be those questions?

Ans. The questions that Ramu might ask Santha are:

1. How these vessels are forming an image like mirrors?
2. All steel vessels form images?
3. Why images are in different shapes and sizes?
4. Why shape and size of images changes as we bring the vessel close to object?
5. Why some images are inverted?

GROUP-B

5. A mixture of Oxygen and Ethyne is burnt for welding. Can you tell why a mixture of Ethyne and air is not used?

Ans: When oxygen and ethyne are burnt it leads to complete burning due to combustion of ethyne and the temperature goes as high as 3000°C , whereas with air this temperature can never be attained due to incomplete burning process.

6. Which method do you suggest for extraction of high reactivity metals? Why?

Ans: We can use Electrolysis of fused compounds. We could have used chemical reduction but due to high temperature requirement this method is not viable.

7. Write any two uses and two properties of Covalent compounds

Ans: Uses of Covalent Compounds –

- These are used in soft drinks. Infact most of the food are made up from covalent bonds.
- Some of them are also used in pesticides and soaps and detergents.

Properties of Covalent Compounds

- They have low boiling and melting points.
- They are poor conductor of electricity.
- They are brittle solids.

8. Where do we use washing method in our daily life? Give example. How do you correlate this example with enrichment of ore?

Ans: In our daily life we use washing for utensils, clothes etc. Ore is the impure form of the metal. We do enrichment of ore to get the pure metal. The process of removing the gangue from the ore is based on the physical and chemical properties of the ore. The ore is first crushed and then the physical process like hydraulic washing, froth floatation and magnetic separation is carried out to remove impurities which are similar to washing of the daily life.

SECTION- II

4x1=4

Note:1. Answer **ANY FOUR** questions from the following.

2. Each question carries **ONE** mark.

9.What is Humidity?

Ans: There are water vapours present in the atmosphere that we called as humidity.

10. Define Critical angle.

Ans: It is defined as angle of incidence where the angle of refraction is 90° .

11.What is the reason for using Tungsten as a filament in electric bulb?

Ans: Due to its high melting point it is normally used as a filament in electric bulb.

12.What is a Chemical bond?

Ans: Any kind of attraction between ions, molecules and atoms and which leads to formation of chemical compounds are known as chemical bonds.

13. Name the simplest Hydrocarbon.

Ans: The simplest Hydrocarbon is methane, CH_4 . This is the simplest member of a series of hydrocarbons.

14. Mention two methods, which produce very pure metals.

Ans: Electrolytic reduction, smelting are two methods which produce very pure metals.

SECTION – III**4x4=16****Note:**

1. Answer **ANY FOUR** questions, choosing at least two from each Group.
2. Each question carries **FOUR** marks.

GROUP -A

15. Explain, why dogs pant during hot summer days using the concept of evaporation?

Ans: Dogs have tongues with large surface area. The water molecules which are present on the surface of the tongue of a dog gets evaporated while panting. This evaporation cools the area. Hence, dogs pant during hot summer days as panting promotes cooling effect and relieves them from the heat.

16. How can you verify that a current carrying wire produces a magnetic field with the help of an experiment?

Ans: If some needles are placed along the wire carrying current and it shows some movement then this proves that it has produced a magnetic field.

17. Write about your favourite scientist / physicist.

Ans: Let us try to know something about Sir Isaac Newton. He is known for its phenomenal contribution in the field of science.

- Known for its newton's law of motions.
- Has given the law of gravitation.
- His contribution is known in the field of Physics, Maths, Astronomy and Economics.

18. A house has four tube lights, three fans and a television. Each tube light draws 40 W. The fan draws 80 W and the television draws 100 W. On an average, all the tube lights are kept on for 5 hours, all fans for 12 hours and the television for 6 hours every day. Find the cost of electric energy used in 30 days at the rate of Rs. 3.00 per KWH.

Ans:

1 tube light consumes power = 40 W

Therefore, 4 tube lights consume power = $4 \times 40 \text{ W} = 160 \text{ W}$

4 tube lights are kept on for five hours. So, consumption of power by 4 tube lights. = $5 \times 160 \text{ W} = 800 \text{ W}$ -----(1)

1 fan consumes power = 80 W

Therefore, 3 fans consume power = $3 \times 80 \text{ W} = 240 \text{ W}$ 3 fans are kept on for 12 hours. So, consumption of power by 3 fans. = $12 \times 240 \text{ W} = 2880 \text{ W}$ ------(2)

TV draws power = 100 W

TV is kept on for 6 hours = $6 \times 100 \text{ W} = 600 \text{ W}$ ------(3)

Therefore, In One day consumption of power = (1) + (2) + (3)
= $800 \text{ W} + 2880 \text{ W} + 600 \text{ W}$

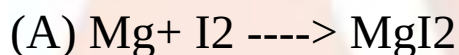
$$= 4280 \text{ W} = 4.280 \text{ KW}$$

Therefore, Total consumption of power in 30 days at Rs. 3 per KW

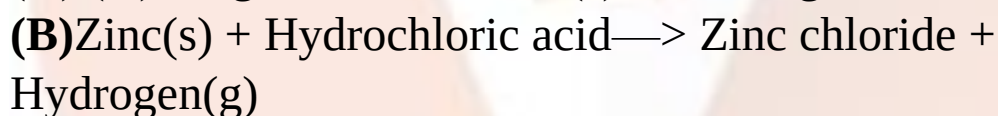
$$= 4.280 \times 30 \times 3 = \text{Rs } 385.2$$

GROUP-B

19. Write the balanced chemical reaction for the following and identify the type of reaction in each case.



Ans:



20. Rainbow is an example for continuous spectrum. Explain.

Ans: Since rainbow is formed when sunlight is dispersed because of water droplets present in atmosphere and it has no sharp boundaries so is called as continuous spectrum

21. What is meant by water of crystallisation of a substance? Describe an activity to show the water of crystallisation.

Ans: The amount of water molecules present inside a substance is known as water of crystallisation of a substance

We can show this by a simple experiment in which we can actually burn copper sulphate crystals and can see the change in color due to removal of water molecules.

Supporting equation - $\text{CuSO}_4 \cdot 5\text{H}_2\text{O} \Rightarrow \text{CuSO}_4$

22. Comment on the position of Hydrogen in Periodic table.

Ans: Hydrogen is the first element of the periodic table with atomic number as 1. It has properties of both alkali as well as halogens since it can lose or gain electrons.

SECTION -IV

1x5=5

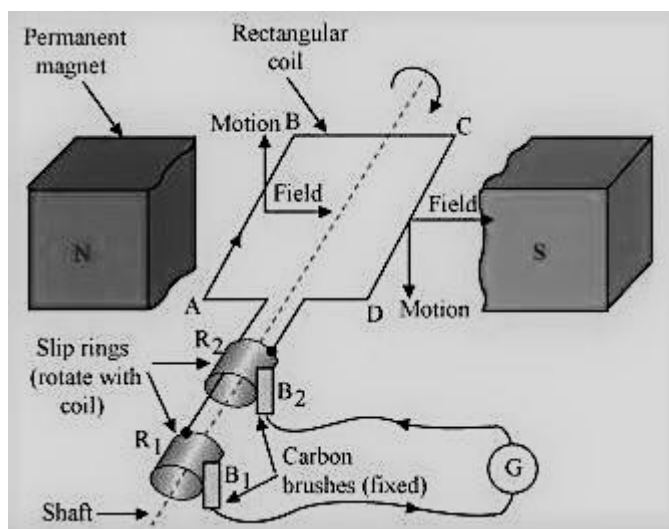
Note:

1. Answer **ANY ONE** of the following question.

2. This question carries **FIVE** marks.

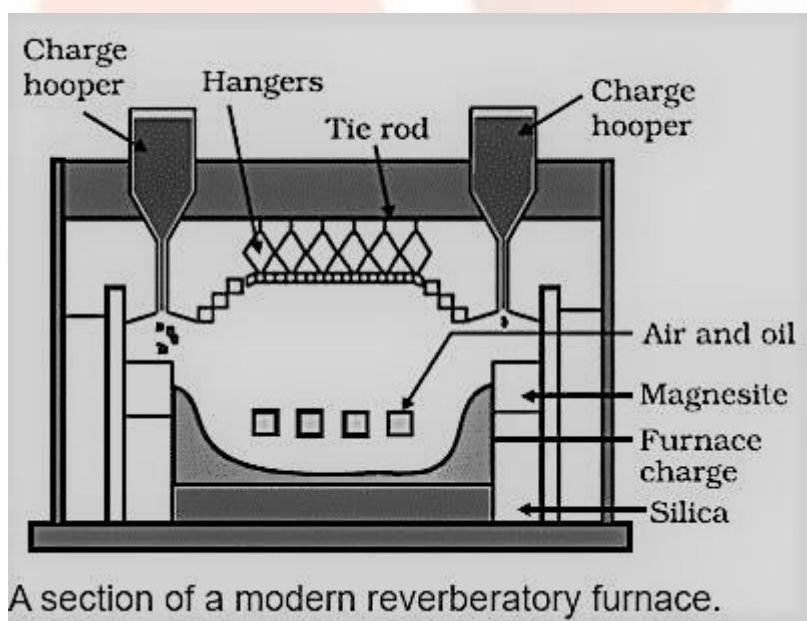
23. Draw a neat diagram of an A.C. generator.

Ans:



24. Draw a neat diagram of Reverberatory furnace and label its parts.

Ans:



Part - B**Time: 30 minutes****Marks: 15**

Write the CAPITAL LETTERS showing the correct answer for the following questions in the brackets provided against them.

20x $\frac{1}{2}$ =10

1. Which of the following is a warming process?

[.....]

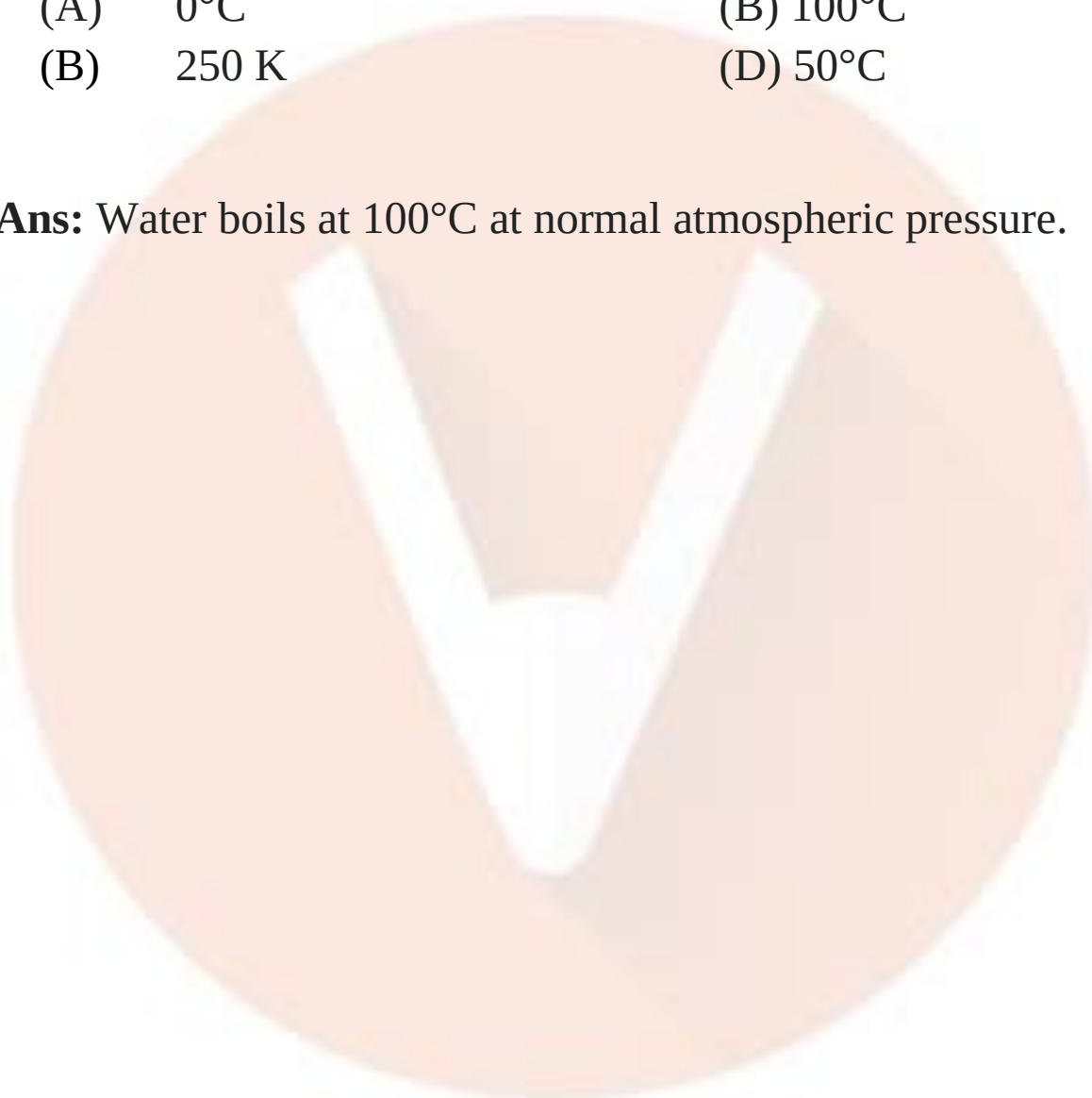
- (A) Evaporation
- (B) Condensation
- (C) Boiling
- (D) All the above.

Ans: Boiling is a warming process.

2. Water at normal atmospheric pressure boils at

- | | |
|-----------|-----------|
| (A) 0°C | (B) 100°C |
| (B) 250 K | (D) 50°C |

Ans: Water boils at 100°C at normal atmospheric pressure.



3. If an object is placed at 'C' on the principal axis in front of a Concave mirror, the position of the image is

- (A) at infinity. (B) between F and C.
(C) at C. (D) beyond C.

Ans: The position of the image will be at C only.

4. The mirror used by E.N.T. specialists is

- (A) Plane (B) Concave
(C) Convex (D) None of these.

Ans: A concave mirror is used by the ENT specialist because it converges the light rays and focuses it on the body parts that are to be examined.

5. Which of the following is Snell's law?

- (A) $n_1 \sin i = \sin r/n_2$ (B) $n_1/n_2 = \sin r/\sin i$
(C) $n_2/n_1 = \sin r/\sin i$ (D) $n_2 \sin i = \text{constant}$

Ans: According to Snell's law $n_1/n_2 = \sin r/\sin i$.

6. The size of an object as perceived by an eye depends primarily on

- (A) Actual size of the object.
- (B) Distance of the object from the eye.
- (C) Aperture of the pupil.
- (D) Size of the image formed on the retina.

Ans: The size of an object as perceived by an eye depends on the size of the image formed on the retina.

7. The current in the wire depends

- (A) Only on the potential difference applied
- (B) Only on the resistance of the wire.
- (C) On potential difference and resistance.
- (D) None of them.

Ans: The current flow in the wire depends on the voltage applied and the resistance of the wire.

8. Which converts Electrical energy into Mechanical energy?

- | | |
|---------------|-------------|
| (A) Motor | (B) Battery |
| (C) Generator | (D) Switch |

Ans: Generator converts Electrical energy into Mechanical energy.

9. Three bodies A, B and C are in thermal equilibrium.

The temperature of B is 45°C . Then the temperature of 'C' is

(A) 45°C

(B) 50°C .

(C) 40°C

(D) any temperature

Ans: Since the three bodies are in thermal equilibrium, the temperature of C will also be 45°C .

10. The value of the focal length of the lens is equal to the value of the image distance, when the rays are

(A) Passing through the Optic centre.

(B) Parallel to the Principal axis.

(C) Passing through the Focus.

(D) In all the above cases.

Ans: When the rays are parallel to the principal axis, the value of the focal length of the lens is equal to the value of the image distance.



The above reaction is an example of

- (A) Chemical combination. (B) Chemical decomposition.
(C) Displacement reaction. (D) Double decomposition reaction.

Ans: The above is an example of a displacement reaction.

12. What happens when dil. HCl is added to iron fillings?

- (A) Hydrogen gas is released.
(B) Chlorine gas is released.
(C) No reaction takes place.
(D) Iron salt and water are produced.

Ans: When dilute hydrochloric acid is added to iron fillings, hydrogen gas and iron chloride are produced.

13. The colour of Methyl orange indicator in acidic medium is

- (A) Yellow (B) Green
(C) Orange (D) Red

Ans - Methyl orange shows red colour in acidic medium.

14. A solution turns red litmus blue. Its pH is likely to be

- | | |
|-------|--------|
| (A) 1 | (B) 4 |
| (C) 5 | (D) 10 |

Ans- Its pH level is likely to be 10.

15. If $l = 1$ for an atom, then the number of orbitals in its subshell is

Ans: It has three orbitals in its subshell.

16. Which of the following is the most active metal?

- | | |
|---------------|--------------|
| (A) Lithium | (B) Sodium |
| (C) Potassium | (D) Rubidium |

Ans: Lithium is the most active metal.

17. Electronic configuration of an atom is 2, 8, 7.

To which of the following elements, would it be chemically similar?

- | | |
|-----------------------------|------------------------|
| (A) Nitrogen ($Z=7$) | (B) Fluorine ($Z=9$) |
| (C) Phosphorus ($Z = 15$) | (D) Argon ($Z=18$) |

Ans: Fluorine is chemically similar to chlorine.

18. An element 'A' forms a chloride ACl_4 .

The number of electrons in the valence shell of 'A' is

- | | |
|-------|-------|
| (A) 1 | (B) 2 |
| (C) 3 | (D) 4 |

Ans: Number of electrons in the valence shell of 'A' is 4

19. Galena is an ore of

- | | |
|------------------|--------------------|
| (A) Zinc (Zn) | (B) Lead (Pb) |
| (C) Mercury (Hg) | (D) Aluminium (Al) |

Ans: Galena also known as lead glance and is a natural form of Lead.

20. 2 ml of ethanoic acid was taken in each of the three test tubes A, B and C, and 2 ml, 4 ml and 8 ml water was added to them respectively, A clear solution is obtained in

- | | |
|------------------------------|------------------------------|
| (A) test tube 'A' only. | (B) test tubes A and B only. |
| (C) test tubes B and C only. | (D) All the test tubes. |

Ans: A clear solution is obtained in all the test tubes.

II. Fill in the blanks with suitable answers.

Each question carries $\frac{1}{2}$ mark.

$$5 \times \frac{1}{2} = 2\frac{1}{2}$$

21. Light selects the least time path to travel between two points. This principle was stated by**Pierre de Fermat**.....

22. Speed of light in vacuum is **186000** m/s.

23. Mirage is an example of **total internal reflection**

24. Myopia can be corrected by using **concave** Lens.



25. Is the symbol of
.....**Resistance**.....

III. Match the following.

26. PCl_3 [.....E.....] (A) 'V' shape.
27. N_2 [...F...] (B) Pyramidal.
28. O_2 [...C.....] (C) Double bond.
29. NH_3 [.....B.....] (D) Tetrahedral.
30. H_2O [...A.....] (E) Trigonal
pyramidal. (F) Triple bond.
(G) Single bond.